

Galerie de la bibliothèque biblioLiaisons3d2d

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Voir le guide pour les détails.

Code :

```
// Parameters :
real R = 2 ;
real L = 3 ;
real theta10 = 55/360*2*pi ;
real theta20 = asin(-R/L*sin(theta10)) ;
real pos = R*cos(theta10) + L *sqrt(1-(R/L*sin(theta10))^2) ;

// Bases et points :
basis b1 = rotationBasis(1, b0, theta10, 'z', b0.z) ;
basis b2 = rotationBasis(2, b0, theta20, 'z', b0.z) ;

triple A = R*b1.x ;
triple B = pos*b0.x ;
real dec = 1 ;
triple C = B + 2*dec*b0.x ;
triple D = B + 4*dec*b0.x ;

// CEC
pen CEC0 = black ;
pen CEC1 = red ;
pen CEC2 = purple ;
pen CEC3 = deepgreen ;
```

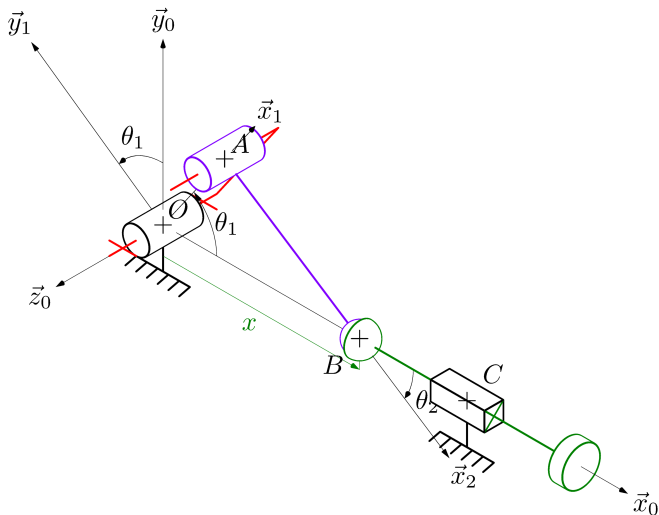
Code :

```
// Relier et bâti
real decBati = 0.75 ;
bati(0-decBati*b0.y, b0.x, -b0.y, CEC0) ;
link(0-decBati*b0.y -- 0, CEC0) ;
bati(C-decBati*b0.y, b0.x, -b0.y, CEC0) ;
link(C-decBati*b0.y -- C, CEC0) ;
real prof = -1 ;
path3 pCEC1 = 0 -- 0+prof*b0.z -- A+prof*b0.z -- A ;
link(pCEC1, CEC1) ;
link(A--B, CEC2) ;
link(B--D, CEC3) ;

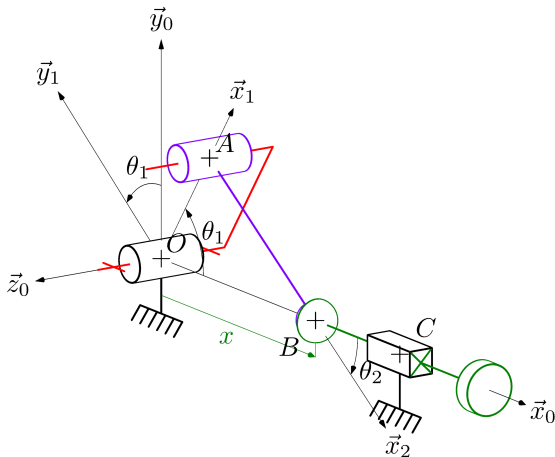
// Liaisons
liaisonPivot(0, b0.z, b0.x, CEC0, CEC1) ;
liaisonPivotGlissant(A, b0.z, CEC2, CEC1) ;
liaisonRotule(B, -b0.x, CEC2, CEC3) ;
liaisonGlissiere(C, b0.x, b0.y, CEC0, CEC3) ;

// Formes supplémentaires
addCylinder(D, b0.x, 0.35, 0.25, CEC3) ;
```

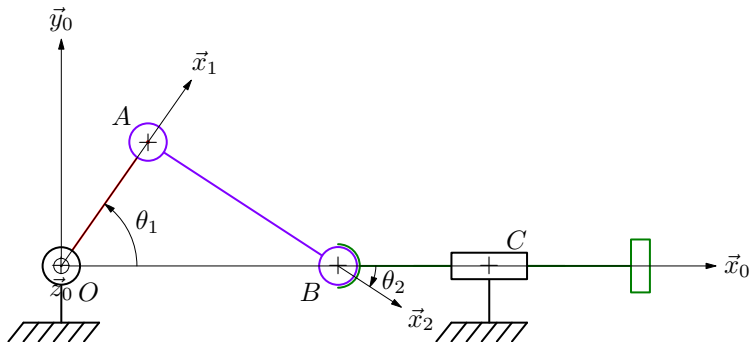
Vue 3D – perspective iso – Paramétrage



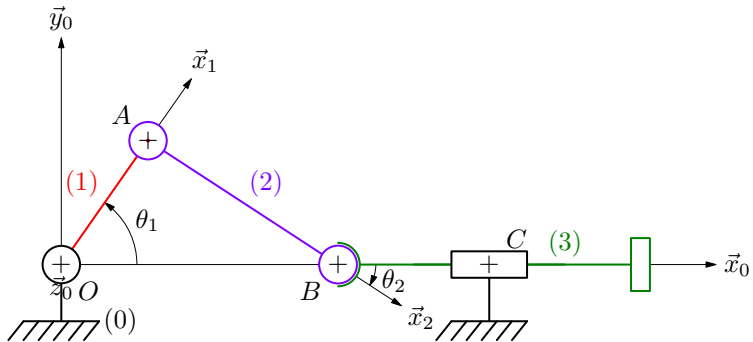
Vue 3D – point de vue (1.5, 0.5, 1) – Code identique



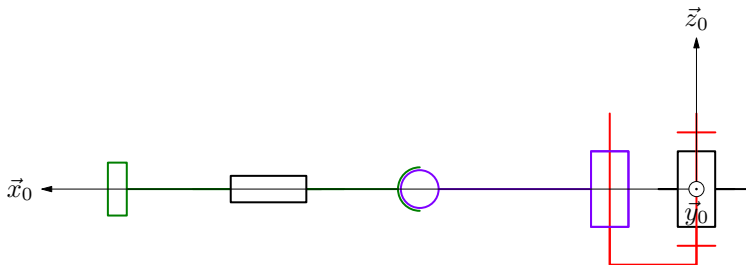
Vue 2D – projection plane – même code !! Le passage 3D 2D est automatique.



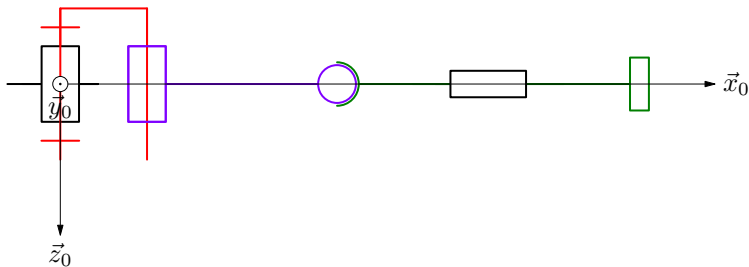
Pour mettre les liaisons au-dessus, il faut simplement inverser l'ordre du code :



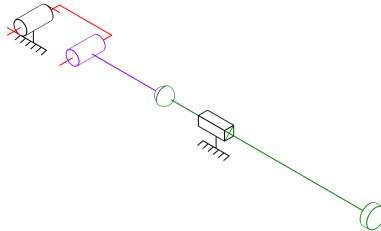
On peut voir dans un autre plan (point de vue $(0, 1, 0)$) avec l'axe $\vec{z}_0$ vers le haut :



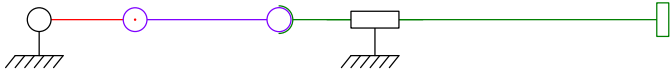
Mais on peut aussi très facilement dire que l'axe vertical est $-\vec{z}_0$,
up devient $(0, 0, -1)$:

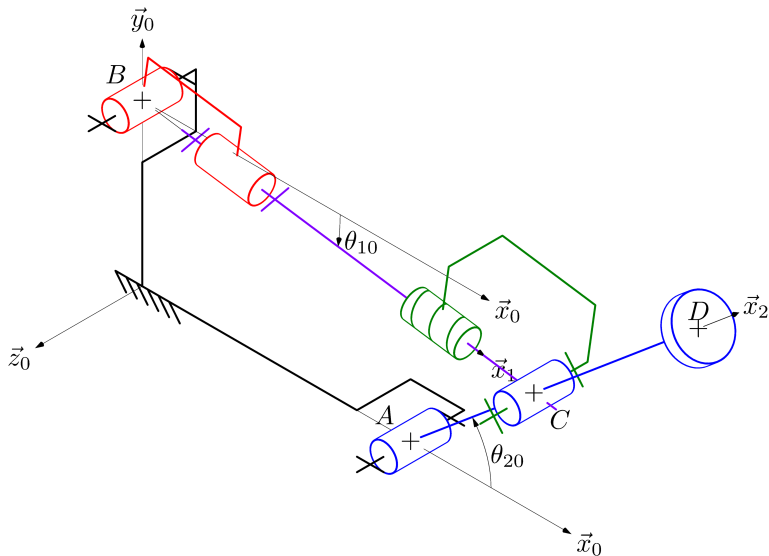


Et on peut animer plus ou moins facilement : OK en 3D (36 points en 4 minutes... qualité moyenne). Cela reste bon néanmoins.

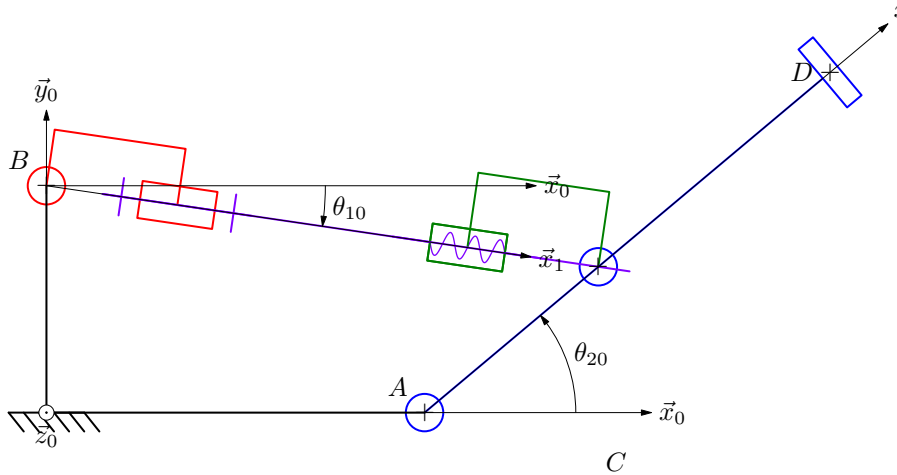


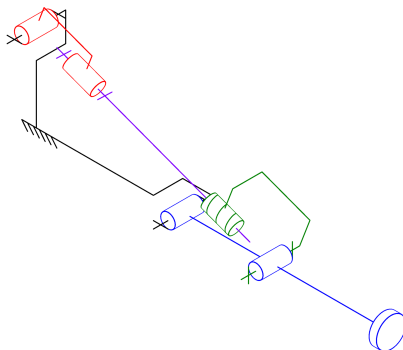
OK en 2D (72 points en 20 secondes... c'est du vectoriel donc léger).



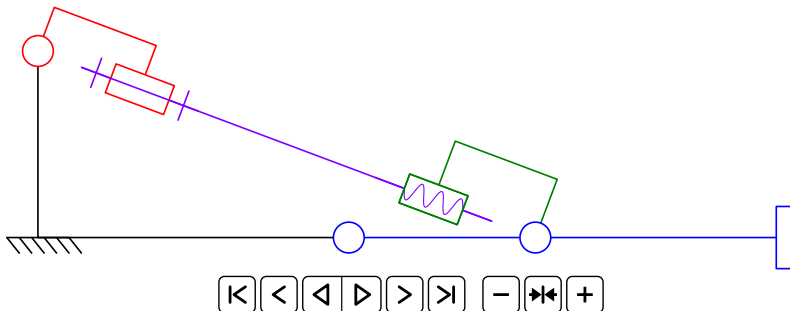




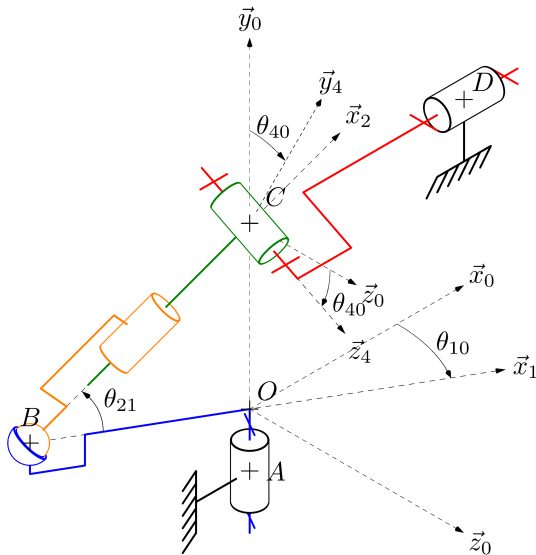




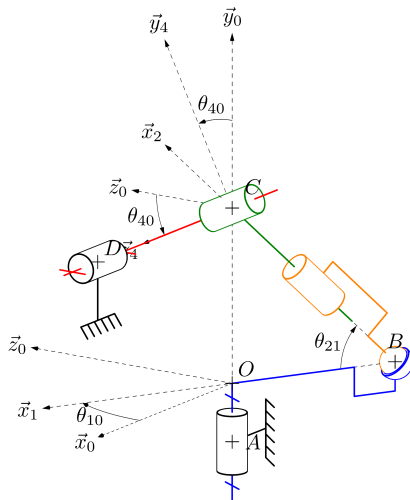
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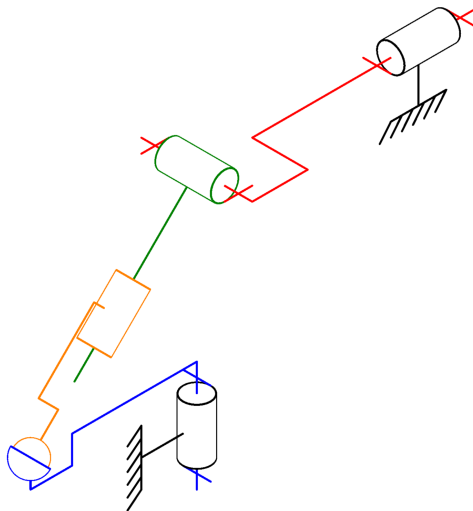


Vue 3D – perspective iso – Paramétrage

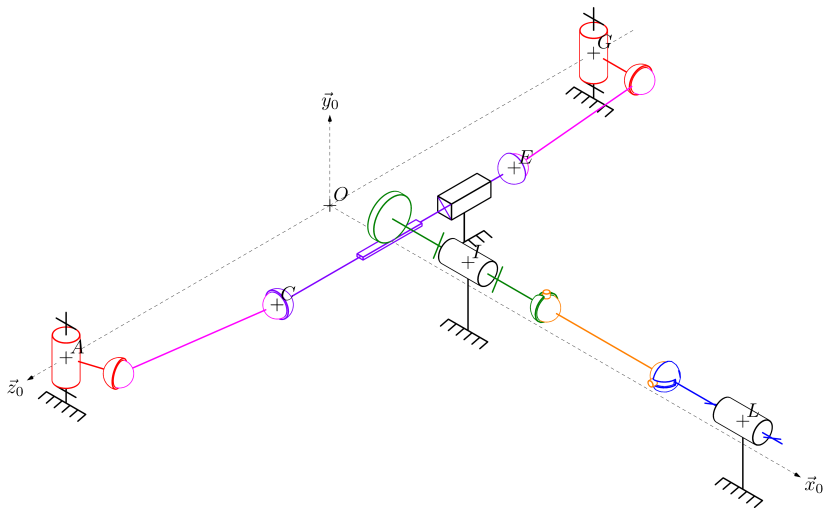


Autre vue

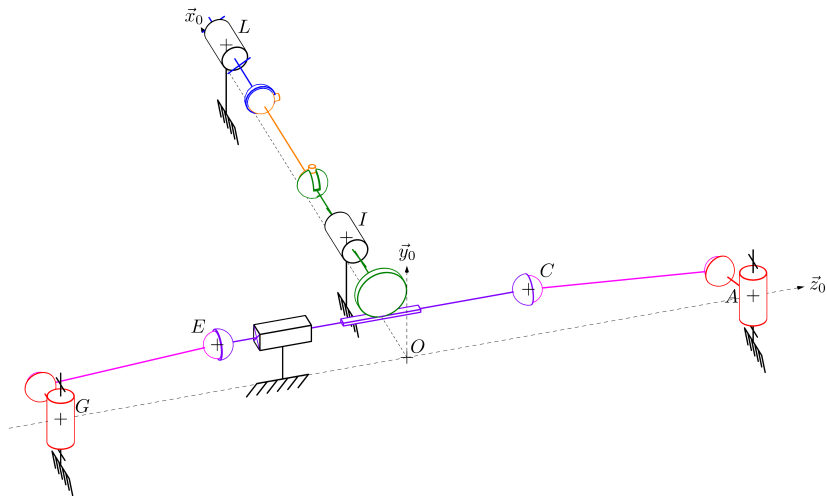


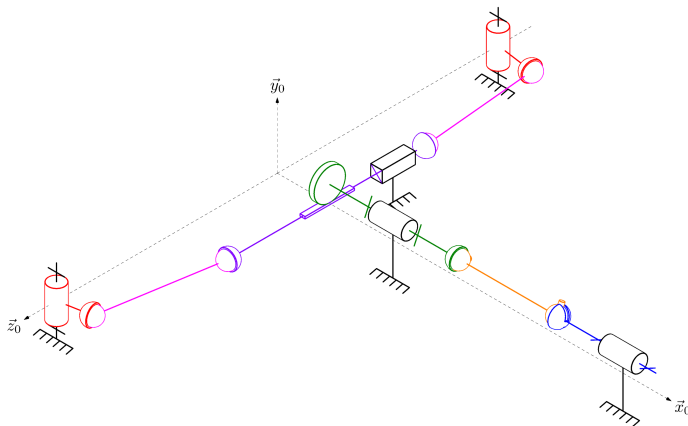


Vue 3D – perspective iso – Paramétrage

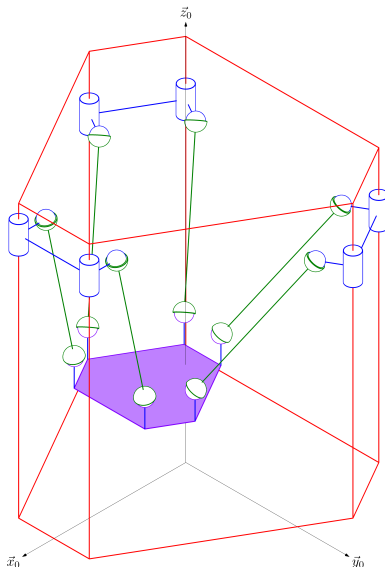


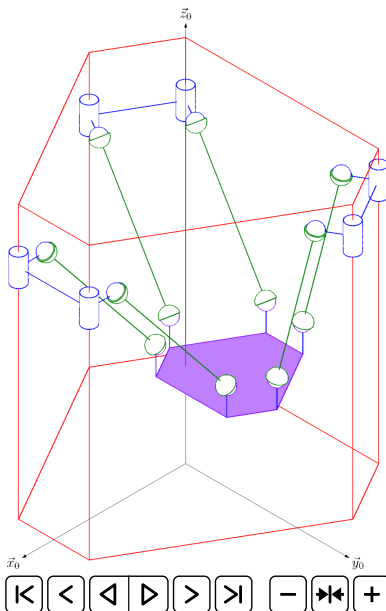
Autre vue





Vue 3D – perspective iso





Vue 3D – perspective iso

